# Manage key managers

ONTAP 9.7 REST API reference

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# Manage key managers

## Security key-managers endpoint overview

### Overview

A key manager is a key management solution (software or dedicated hardware) that enables other ONTAP client modules to securely and persistently store keys for various uses. For example, WAFL uses the key management framework to store and retrieve the volume encryption keys that it uses to encrypt/decrypt data on NVE volumes. A key manager can be configured at both cluster scope and SVM, with one key manager allowed per SVM. The key management framework in ONTAP supports two mutually exclusive modes for persisting keys: external and onboard.

When an SVM is configured with external key management, the keys are stored on up to four key servers that are external to the system.

Once external key management is enabled for an SVM, key servers can be added or removed using the */api/security/key-managers/{uuid}/key-servers* endpoint. See [`POST /security/key-managers/{uuid}/key-servers`] and [`DELETE /security/key-managers/{uuid}/key-servers/{server}`] for more details.

Setting up external key management dictates that the required certificates for securely communicating with the key server are installed prior to configuring the key manager. To install the required client and server_ca certificates, use the */api/security/certificates/* endpoint.

See [`POST /security/certificates`], [`GET /security/certificates/uuid`] and [`DELETE /security/certificates/{uuid}`] for more details.

When an SVM is configured with onboard key management, the keys are stored in ONTAP in wrapped format using a key hierarchy created using the salted hash of the passphrase entered when configuring onboard key management. This model fits well for customers who use ONTAP to store their own data.

### Examples

#### Creating an external key manager with 1 key server for a cluster

The example key manager is configured at the cluster-scope with one key server. Note that the UUIDs of the certificates are those that are already installed at the cluster-scope. Note the *return_records=true* query parameter is used to obtain the newly created key manager configuration

```
# The API:
POST /api/security/key-managers

# The call:
curl -X POST 'https://<mgmt-ip>/api/security/key-
managers?return_records=true' -H 'accept: application/hal+json' -d "{
\"external\": { \"client_certificate\": { \"uuid\": \"5fb1701a-d922-11e8-
bfe8-005056bb017d\" }, \"server_ca_certificates\": [ { \"uuid\":
\"827d7d31-d6c8-11e8-b5bf-005056bb017d\" }],\"servers\": [ { \"server\":
\"10.225.89.33:5696\" } ] } }"

# The response:
{
"num_records": 1,
"records": [
  {
    "uuid": "815e9462-dc57-11e8-9b2c-005056bb017d",
    "external": {
      "client_certificate": {
        "uuid": "5fb1701a-d922-11e8-bfe8-005056bb017d"
      },
      "server_ca_certificates": [
        {
          "uuid": "827d7d31-d6c8-11e8-b5bf-005056bb017d"
        }
      ],
      "servers": [
        {
          "server": "10.225.89.33:5696"
        }
      ]
    },
    "_links": {
      "self": {
        "href": "/api/security/key-managers/815e9462-dc57-11e8-9b2c-
005056bb017d"
      }
    }
  }
]
}
```

### Creating an external key manager with 1 key server for an SVM

The example key manager is configured at the SVM-scope with one key server. Note that the UUIDs of the certificates are those that are already installed in that SVM. Note the *return_records=true* query parameter is used to obtain the newly created key manager configuration

```
# The API:
POST /api/security/key-managers

# The call:
curl -X POST 'https://<mgmt-ip>/api/security/key-
managers?return_records=true' -H 'accept: application/hal+json' -d "{
\"svm\": { \"uuid\": \"216e6c26-d6c6-11e8-b5bf-005056bb017d\" },
\"external\": { \"client_certificate\": { \"uuid\": \"91dcaf7c-dbbd-11e8-
9b2c-005056bb017d\" }, \"server_ca_certificates\": [ { \"uuid\":
\"a4d4b8ba-dbbd-11e8-9b2c-005056bb017d\" }],\"servers\": [ { \"server\":
\"10.225.89.34:5696\" } ] } }"

# The response:
{
"num_records": 1,
"records": [
  {
    "uuid": "80af63f2-dbbf-11e8-9b2c-005056bb017d",
    "svm": {
      "uuid": "216e6c26-d6c6-11e8-b5bf-005056bb017d"
    },
    "external": {
      "client_certificate": {
      "uuid": "91dcaf7c-dbbd-11e8-9b2c-005056bb017d"
      },
      "server_ca_certificates": [
        {
          "uuid": "a4d4b8ba-dbbd-11e8-9b2c-005056bb017d"
        }
      ],
      "servers": [
        {
          "server": "10.225.89.34:5696"
        }
      ]
    },
    "_links": {
      "self": {
        "href": "/api/security/key-managers/80af63f2-dbbf-11e8-9b2c-
005056bb017d"
      }
    }
  }
]
}
```

#### Creating an onboard key manager for a cluster

The following example shows how to create an onboard key manager for a cluster with the onboard key manager configured at the cluster-scope.

```
# The API:
POST /api/security/key-managers

# The call:
curl -X POST 'https://<mgmt-ip>/api/security/key-managers' -H 'accept:
application/hal+json' -d '{ "onboard": { "passphrase": "passphrase" } }'
```

#### Retrieving the key manager configurations for all clusters and SVMs

The following example shows how to retrieve all configured key managers along with their configurations.

```
# The API:
GET /api/security/key-managers

# The call:
curl -X GET 'https://<mgmt-ip>/api/security/key-managers?fields=*' -H
'accept: application/hal+json'

# The response:
{
"records": [
  {
    "uuid": "2345f09c-d6c9-11e8-b5bf-005056bb017d",
    "scope": "svm",
    "svm": {
      "uuid": "0f22f8f3-d6c6-11e8-b5bf-005056bb017d",
      "name": "vs0"
    },
    "external": {
      "client_certificate": {
        "uuid": "4cb15482-d6c8-11e8-b5bf-005056bb017d",
        "_links": {
          "self": {
            "href": "/api/security/certificates/4cb15482-d6c8-11e8-b5bf-
005056bb017d/"
          }
        }
      },
      "server_ca_certificates": [
```

```
        {
          "uuid": "8a17c858-d6c8-11e8-b5bf-005056bb017d",
          "_links": {
            "self": {
              "href": "/api/security/certificates/8a17c858-d6c8-11e8-b5bf-
005056bb017d/"
            }
          }
        }
      ],
      "servers": [
        {
          "server": "10.2.30.4:5696",
          "timeout": 25,
          "username": "",
          "_links": {
            "self": {
              "href": "/api/security/key-managers/2345f09c-d6c9-11e8-b5bf-
005056bb017d/key-servers/10.2.30.4:5696/"
            }
          }
        },
        {
          "server": "vs0.local1:3678",
          "timeout": 25,
          "username": "",
          "_links": {
            "self": {
              "href": "/api/security/key-managers/2345f09c-d6c9-11e8-b5bf-
005056bb017d/key-servers/vs0.local1:3678/"
            }
          }
        }
      ]
    },
    "_links": {
      "self": {
        "href": "/api/security/key-managers/2345f09c-d6c9-11e8-b5bf-
005056bb017d"
      }
    }
  },
  {
    "uuid": "815e9462-dc57-11e8-9b2c-005056bb017d",
    "scope": "cluster",
    "external": {
```

```
      "client_certificate": {
        "uuid": "5fb1701a-d922-11e8-bfe8-005056bb017d",
        "_links": {
          "self": {
            "href": "/api/security/certificates/5fb1701a-d922-11e8-bfe8-
005056bb017d/"
          }
        }
      },
      "server_ca_certificates": [
        {
          "uuid": "827d7d31-d6c8-11e8-b5bf-005056bb017d",
          "_links": {
            "self": {
              "href": "/api/security/certificates/827d7d31-d6c8-11e8-b5bf-
005056bb017d/"
            }
          }
        }
      ],
      "servers": [
        {
          "server": "10.225.89.33:5696",
          "timeout": 25,
          "username": "",
          "_links": {
            "self": {
              "href": "/api/security/key-managers/815e9462-dc57-11e8-9b2c-
005056bb017d/key-servers/10.225.89.33:5696/"
            }
          }
        }
      ]
    },
    "_links": {
      "self": {
        "href": "/api/security/key-managers/815e9462-dc57-11e8-9b2c-
005056bb017d"
      }
    }
  }
],
"num_records": 2,
"_links": {
  "self": {
    "href": "/api/security/key-managers?fields=*"
```

```
  }
}
}
```

---

**Retrieving the key manager configurations for all clusters and SVMs (showing Onboard Key Manager)**

The following example shows how to retrieve all configured key managers along with their configurations.

```
# The API:
GET /api/security/key-managers

# The call:
curl -X GET 'https://<mgmt-ip>/api/security/key-managers?fields=*' -H
'accept: application/hal+json'

# The response:
{
"records": [
  {
    "uuid": "8ba52e0f-ae22-11e9-b747-005056bb7636",
    "scope": "cluster",
    "onboard": {
      "enabled": true,
      "key_backup": "--------------------------BEGIN
BACKUP--------------------------\n <Backup Data>
\n--------------------------END BACKUP--------------------------\n"
    },
    "volume_encryption": {
      "supported": false,
      "message": "The following nodes do not support volume granular
encryption: ntap-vsim2.",
      "code": 65536935
    },
    "is_default_data_at_rest_encryption_disabled": false
  }
],
"num_records": 1
}
```

---

**Retrieving expensive fields such as, status.code and status.message, associated with a key manager.**

These values are not retreived by default with the 'fields=*' option. The following example shows how to

retrieve the expensive objects associated with a key manager.

```
# The API:
GET /api/security/key-managers

# The call:
curl -X GET "https://<mgmt-ip>/api/security/key-
managers?fields=status.message",status.code" -H 'accpt:
application/hal+jon'

# The response:
{
"records": [
  {
    "uuid": "ac305d46-aef4-11e9-ad3c-005056bb7636",
    "status": {
      "message": "No action needed at this time.",
      "code": 65537200
    },
    "_links": {
      "self": {
        "href": "/api/security/key-managers/ac305d46-aef4-11e9-ad3c-
005056bb7636"
      }
    }
  }
],
"num_records": 1,
"_links": {
  "self": {
    "href": "/api/security/key-managers?fields=status.message,status.code"
  }
}
```

---

#### Retrieving a specific key manager configuration

The following example shows how to retrieve a specific key manager configuration.

```
# The API:
GET /api/security/key-managers/{uuid}

# The call:
curl -X GET 'https://<mgmt-ip>/api/security/key-managers/<uuid>?fields=*'
-H 'accept: application/hal+json'
```

---

```
# The response:
{
"uuid": "2345f09c-d6c9-11e8-b5bf-005056bb017d",
"scope": "svm",
"svm": {
  "uuid": "0f22f8f3-d6c6-11e8-b5bf-005056bb017d",
  "name": "vs0"
},
"external": {
  "client_certificate": {
    "uuid": "4cb15482-d6c8-11e8-b5bf-005056bb017d",
    "_links": {
      "self": {
        "href": "/api/security/certificates/4cb15482-d6c8-11e8-b5bf-
005056bb017d/"
      }
    }
  },
  "server_ca_certificates": [
    {
      "uuid": "8a17c858-d6c8-11e8-b5bf-005056bb017d",
      "_links": {
        "self": {
          "href": "/api/security/certificates/8a17c858-d6c8-11e8-b5bf-
005056bb017d/"
        }
      }
    }
  ],
  "servers": [
    {
      "server": "10.2.30.4:5696",
      "timeout": 25,
      "username": "",
      "_links": {
        "self": {
          "href": "/api/security/key-managers/2345f09c-d6c9-11e8-b5bf-
005056bb017d/key-servers/10.2.30.4:5696/"
        }
      }
    },
    {
      "server": "vs0.local1:3678",
      "timeout": 25,
      "username": "",
```

```
      "_links": {
        "self": {
          "href": "/api/security/key-managers/2345f09c-d6c9-11e8-b5bf-
005056bb017d/key-servers/vs0.local1:3678/"
        }
      }
    }
  ]
},
"_links": {
  "self": {
    "href": "/api/security/key-managers/2345f09c-d6c9-11e8-b5bf-
005056bb017d"
  }
}
}
```

#### Updating the configuration of an external key manager

The following example shows how to update the server-ca configuration of an external key manager.

```
# The API:
PATCH /api/security/key-managers/{uuid}

# The call:
curl -X PATCH 'https://<mgmt-ip>/api/security/key-managers/<uuid>?' -H
'accept: application/hal+json' -d "{ \"external\": {
\"server_ca_certificates\": [ { \"uuid\": \"23b05c58-d790-11e8-b5bf-
005056bb017d\" }] } }"
```

#### Updating the passphrase of an Onboard Key Manager

The following example shows how to update the passphrase of a given key manager.

```
# The API:
PATCH /api/security/key-managers/{uuid}

# The call:
curl -X PATCH 'https://<mgmt-ip>/api/security/key-managers/<uuid>?' -H
'accept: application/hal+json' -d "{ \"onboard\": {
\"existing_passphrase\": \"existing_passphrase\", \"passphrase\":
\"new_passphrase\" } }"
```

#### Synchronizing the passphrase of the Onboard Key Manager on a cluster

The following example shows how to synchronize the passphrase on a cluster where the Onboard Key Manager is already configured.

```
# The API:
PATCH /api/security/key-managers/{uuid}

# The call:
curl -X PATCH 'https://<mgmt-ip>/api/security/key-managers/<uuid>?' -H
'accept: application/hal+json' -d "{  \"onboard\": {
\"existing_passphrase\": \"existing_passphrase\",    \"synchronize\": true
}}"
```

#### Configuring the Onboard Key Manager on a cluster

The following example shows how to configure the Onboard Key Manager on a cluster where the Onboard Key Manager is not configured, but is configured on an MetroCluster partner cluster.

```
# The API:
POST /api/security/key-managers

# The call:
curl -X POST 'https://<mgmt-ip>/api/security/key-
managers?return_records=false' -H  'accept: application/hal+json' -H
"Content-Type: application/json" -d "{  \"onboard\": {    \"passphrase\":
\"passphrase\",    \"synchronize\": true  }}"
```

#### Deleting a configured key manager

The following example shows how to delete a key manager given its UUID.

```
# The API:
DELETE /api/security/key-managers/{uuid}

# The call:
curl -X DELETE 'https://<mgmt-ip>/api/security/key-managers/<uuid>?' -H
'accept: application/hal+json'
```

---

#### Adding a key server to an external key manager

The following example shows how to add a key server to an external key manager.

```
# The API:
POST /api/security/key-managers/{uuid}/key-servers

# The call:
curl -X POST 'https://<mgmt-ip>/api/security/key-managers/<uuid>/key-
servers?return_records=true' -H 'accept: application/hal+json' -d "{
\"server\": \"10.225.89.34:5696\" }"

# The response:
{
"num_records": 1,
"records": [
  {
    "server": "10.225.89.34:5696",
    "_links": {
      "self": {
        "href": "/api/security/key-managers/43e0c191-dc5c-11e8-9b2c-
005056bb017d/key-servers/10.225.89.34%3A5696"
      }
    }
  }
]
}
```

---

#### Adding 2 key servers to an external key manager

The following example shows how to add 2 key servers to an external key manager. Note that the *records* property is used to add multiple key servers to the key manager in a single API call.

---

```
# The API:
POST /api/security/key-managers/{uuid}/key-servers

# The call:
curl -X POST 'https://<mgmt-ip>/api/security/key-managers/<uuid>/key-
servers?return_records=true' -H 'accept: application/hal+json' -d "{
\"records\": [ { \"server\": \"10.225.89.34:5696\" }, { \"server\":
\"10.225.89.33:5696\" } ] }"

# The response:
{
"num_records": 1,
"records": [
  {
    "_links": {
      "self": {
        "href": "/api/security/key-managers/43e0c191-dc5c-11e8-9b2c-
005056bb017d/key-servers/"
      }
    }
  }
]
}
```

---

**Retrieving all the key servers configured in an external key manager**

The following example shows how to retrieve all key servers configured in an external key manager.

```
# The API:
GET /api/security/key-managers/{uuid}/key-servers

# The call:
curl -X GET 'https://<mgmt-ip>/api/security/key-managers/<uuid>/key-
servers?fields=*' -H 'accept: application/hal+json'

# The response:
{
"records": [
  {
    "uuid": "43e0c191-dc5c-11e8-9b2c-005056bb017d",
    "server": "10.225.89.33:5696",
    "timeout": 25,
    "username": "",
    "_links": {
      "self": {
        "href": "/api/security/key-managers/43e0c191-dc5c-11e8-9b2c-
005056bb017d/key-servers/10.225.89.33%3A5696"
      }
    }
  },
  {
    "uuid": "43e0c191-dc5c-11e8-9b2c-005056bb017d",
    "server": "10.225.89.34:5696",
    "timeout": 25,
    "username": "",
    "_links": {
      "self": {
        "href": "/api/security/key-managers/43e0c191-dc5c-11e8-9b2c-
005056bb017d/key-servers/10.225.89.34%3A5696"
      }
    }
  }
],
"num_records": 2,
"_links": {
  "self": {
    "href": "/api/security/key-managers/43e0c191-dc5c-11e8-9b2c-
005056bb017d/key-servers?fields=*"
  }
}
}
```

#### Retrieving a specific key server configured in an external key manager

The following example shows how to retrieve a specific key server configured in an external key manager.

```
# The API:
GET /api/security/key-managers/{uuid}/key-servers/{server}

# The call:
curl -X GET 'https://<mgmt-ip>/api/security/key-managers/<uuid>/key-
servers/{server}?fields=*' -H 'accept: application/hal+json'

# The response:
{
"uuid": "43e0c191-dc5c-11e8-9b2c-005056bb017d",
"server": "10.225.89.34:5696",
"timeout": 25,
"username": "",
"_links": {
  "self": {
    "href": "/api/security/key-managers/43e0c191-dc5c-11e8-9b2c-
005056bb017d/key-servers/10.225.89.34:5696"
  }
}
}
```

#### Updating a specific key server configuration configured in an external key manager

The following example shows how to update a specific key server configured in an external key manager.

```
# The API:
PATCH /api/security/key-managers/{uuid}/key-servers/{server}

# The call:
curl -X PATCH 'https://<mgmt-ip>/api/security/key-managers/<uuid>/key-
servers/{server}' -H 'accept: application/hal+json' -d "{ \"timeout\": 45
}"
```

#### Deleting a key server from an external key manager

The following example shows how to delete a key server from an external key manager.

```
# The API:
DELETE /api/security/key-managers/{uuid}/key-servers/{server}

# The call:
curl -X DELETE 'https://<mgmt-ip>/api/security/key-managers/<uuid>/key-
servers/{server}' -H 'accept: application/hal+json'
```

# Retrieve key managers

GET `/security/key-managers`

Retrieves key managers.

## Expensive properties

There is an added cost to retrieving values for these properties. They are not included by default in GET results and must be explicitly requested using the `fields` query parameter. See DOC Requesting specific fields to learn more.

- `connectivity`
- `status.message`
- `status.code`

## Related ONTAP commands

- `security key-manager show-keystore`
- `security key-manager external show`
- `security key-manager external show-status`
- `security key-manager onboard show-backup`

## Learn more

- DOC /security/key-managers

## Parameters

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| svm.uuid | string | query | False | Filter by svm.uuid |
| svm.name | string | query | False | Filter by svm.name |
| uuid | string | query | False | Filter by uuid |

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| volume_encryption.code | integer | query | False | Filter by volume_encryption.code |
| volume_encryption.supported | boolean | query | False | Filter by volume_encryption.supported |
| volume_encryption.message | string | query | False | Filter by volume_encryption.message |
| is_default_data_at_rest_encryption_disabled | boolean | query | False | Filter by is_default_data_at_rest_encryption_disabled |
| external.servers.username | string | query | False | Filter by external.servers.username |
| external.servers.server | string | query | False | Filter by external.servers.server |
| external.servers.connectivity.cluster_availability | boolean | query | False | Filter by external.servers.connectivity.cluster_availability |
| external.servers.connectivity.records.state | string | query | False | Filter by external.servers.connectivity.records.state |
| external.servers.connectivity.records.node.uuid | string | query | False | Filter by external.servers.connectivity.records.node.uuid |
| external.servers.connectivity.records.node.name | string | query | False | Filter by external.servers.connectivity.records.node.name |

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| external.servers.timeout | integer | query | False | Filter by external.servers.timeout |
| external.server_ca_certificates.uuid | string | query | False | Filter by external.server_ca_certificates.uuid |
| external.client_certificate.uuid | string | query | False | Filter by external.client_certificate.uuid |
| onboard.enabled | boolean | query | False | Filter by onboard.enabled |
| onboard.key_backup | string | query | False | Filter by onboard.key_backup |
| status.message | string | query | False | Filter by status.message |
| status.code | integer | query | False | Filter by status.code |
| scope | string | query | False | Filter by scope |
| fields | array[string] | query | False | Specify the fields to return. |
| max_records | integer | query | False | Limit the number of records returned. |
| return_timeout | integer | query | False | The number of seconds to allow the call to execute before returning. When iterating over a collection, the default is 15 seconds. ONTAP returns earlier if either max records or the end of the collection is reached. |

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| return_records | boolean | query | False | The default is true for GET calls. When set to false, only the number of records is returned. |
| order_by | array[string] | query | False | Order results by specified fields and optional [asc |

## Response

```
Status: 200, Ok
```

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| num_records | integer | Number of records |
| records | array[security_key_manager] | |

**Example response**


```
{
  "_links": {
    "next": {
      "href": "/api/resourcelink"
    },
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "records": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "external": {
      "client_certificate": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
          }
        },
        "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
      },
      "server_ca_certificates": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
          }
        },
        "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
      },
      "servers": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
          }
        },
        "connectivity": {
          "records": {
            "node": {
              "_links": {
                "self": {
                  "href": "/api/resourcelink"
```

```
                }
              },
              "name": "node1",
              "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
            },
            "state": "not_responding"
          }
        },
        "server": "keyserver1.com:5698",
        "timeout": 60,
        "username": "admin"
      }
    },
    "onboard": {
      "existing_passphrase": "The cluster password of length 32-256
ASCII characters.",
      "key_backup": "'--------------------------BEGIN
BACKUP--------------------------
TmV0QXBwIEtleSBCbG9iAAEAAAAEAAAAcAEAAAAAAAAxBFWWAAAAACEAAAAAAAAA
QAAAAAAAAABzDyyVAAAAALI5Jsjvy6gUxnT78KoDKXHYb6sSeraM00quOULY6BeV
n6dMFxuErCD1lbERaOQZSuaYy1p8oQHtTEfGMLZM4TYiAAAAAAAAACgAAAAAAAAA
3WTh7gAAAAAAAAAAAAAAAAIAAAAAAAgAZJEIWvdeHr5RCAvHGclo+wAAAAAAAAAA
IgAAAAAAAAoAAAAAAAAAEOTcR0AAAAAAAAAAAAAAAACAAAAAAAJAGr3tJA/LRzU
QRHwv+1aWvAAAAAAAAAACQAAAAAAAAgAAAAAAAADV1Vd/AAAAAMFM9Q229Bhp
mDaTSdqku5DCd8wG+fOZSr4bx4JT5WHvV/r5gJnXDQQAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAABOZXRBcHAgS2V5IEJsb2IA
AQAAAAMAAAAYAQAAAAAALgePkcAAAAAIgAAAAAAAAAoAAAAAAAAAEOTcR0AAAAA
AAAAAAAAAAACAAAAAAAJAGr3tJA/LRzUQRHwv+1aWvAAAAAAAAAAACIAAAAAAAAA
KAAAAAAAAACIlCHZAAAAAAAAAAAAAAAAAgAAAAAAAQCafcabsxRXMM7gxhLRrzxh
AAAAAAAAAAAkAAAAAAAAAIAAAAAAAAAA2JjQBQAAAACt4IqXcNpVggahl0axLsN4
yQjnNVKWY7mANB29O42hI7b70DTGCTaVAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAE5ldEFwcCBLZXkgQmxvYgABAAAAAwAAABgBAAAAAAAA
7sbaoQAAAAAiAAAAAAAAACgAAAAAAAAQ5NxHQAAAAAAAAAAAAAAAAAIAAAAAAAkA
ave0kD8tHNRBEfC/7Vpa8AAAAAAAAAAAIgAAAAAAAAoAAAAAAAAALOHfWkAAAAA
AAAAAAAAAAACAAAAAAABAMoI9UxrHOGthQm/CB+EHdAAAAAAAAAAACQAAAAAAAAA
gAAAAAAAAACnMmUtAAAAAGVk8AtPzENFgsGdsFvnmucmYrlQCsFew0HDSFKaZqK6
W8IEVzBAhPoAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
```

```
----------------------------END BACKUP----------------------------'",
      "passphrase": "The cluster password of length 32-256 ASCII
characters."
    },
    "scope": "svm",
    "status": {
      "code": 346758,
      "message": "This cluster is part of a MetroCluster configuration.
Use the REST API POST method security/key_managers/ with the
synchronize option and the same passphrase on the partner cluster
before proceeding with any key manager operations.  Failure to do so
could lead to switchover or switchback failure."
    },
    "svm": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "svm1",
      "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
    },
    "uuid": "string",
    "volume_encryption": {
      "code": 346758,
      "message": "No platform support for volume encryption in
following nodes - node1, node2."
    }
  }
}
```


## Error

```
Status: Default, Error
```

| Name | Type | Description |
|---|---|---|
| error | error | |

**Example error**

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

## Definitions

**See Definitions**

href

| Name | Type | Description |
|---|---|---|
| href | string | |

_links

| Name | Type | Description |
|---|---|---|
| next | href | |
| self | href | |

_links

| Name | Type | Description |
|---|---|---|
| self | href | |

client_certificate

Client certificate

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| uuid | string | Certificate UUID |

server_ca_certificates

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| uuid | string | Certificate UUID |

self_link

| Name | Type | Description |
|---|---|---|
| self | href | |

node

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| name | string | |

| Name | Type | Description |
|---|---|---|
| uuid | string | |

key_server_state

The state of the key server for a specific node.

| Name | Type | Description |
|---|---|---|
| node | node | |
| state | string | Key server connectivity state |

connectivity

This property returns the key server connectivity state on all nodes of the cluster. The state is returned for a node only if the connectivity is not in an available state on that node. This is an advanced property; there is an added cost to retrieving its value. The property is not populated for either a collection GET or an instance GET unless it is explicitly requested using the `fields` query parameter or GET for all advanced properties is enabled.

| Name | Type | Description |
|---|---|---|
| cluster_availability | boolean | Set to true when key server connectivity state is available on all nodes of the cluster. |
| records | array[key_server_state] | An array of key server connectivity states for each node. |

key_server_readcreate

| Name | Type | Description |
|---|---|---|
| _links | self_link | |
| connectivity | connectivity | This property returns the key server connectivity state on all nodes of the cluster. The state is returned for a node only if the connectivity is not in an available state on that node. This is an advanced property; there is an added cost to retrieving its value. The property is not populated for either a collection GET or an instance GET unless it is explicitly requested using the `fields` query parameter or GET for all advanced properties is enabled. |

| Name | Type | Description |
| --- | --- | --- |
| server | string | External key server for key management. If no port is provided, a default port of 5696 is used. |
| timeout | integer | I/O timeout in seconds for communicating with the key server. |
| username | string | Username credentials for connecting with the key server. |

external

Configures external key management

| Name | Type | Description |
| --- | --- | --- |
| client_certificate | client_certificate | Client certificate |
| server_ca_certificates | array[server_ca_certificates] | The UUIDs of the server CA certificates already installed in the cluster or SVM. The array of certificates are common for all the keyservers per SVM. |
| servers | array[key_server_readcreate] | The set of external key servers. |

onboard

Configures onboard key management. After configuring onboard key management, save the encrypted configuration data in a safe location so that you can use it if you need to perform a manual recovery operation.

| Name | Type | Description |
| --- | --- | --- |
| enabled | boolean | Is the onboard key manager enabled? |
| existing_passphrase | string | The cluster-wide passphrase. This is not audited. |

| Name | Type | Description |
| --- | --- | --- |
| key_backup | string | Backup of the onboard key manager's key hierarchy. It is required to save this backup after configuring the onboard key manager to help in the recovery of the cluster in case of catastrophic failures. |
| passphrase | string | The cluster-wide passphrase. This is not audited. |
| synchronize | boolean | Synchronizes missing onboard keys on any node in the cluster. If a node is added to a cluster that has onboard key management configured, the synchronize operation needs to be performed in a PATCH operation. In a MetroCluster configuration, if onboard key management is enabled on one site, then the synchronize operation needs to be run as a POST operation on the remote site providing the same passphrase. |

status

Optional status information on the current state of the key manager indicating if it is fully setup or requires more action.

| Name | Type | Description |
| --- | --- | --- |
| code | integer | Code corresponding to the status message. Returns 0 if the setup is complete. This is an advanced property; there is an added cost to retrieving its value. The property is not populated for either a collection GET or an instance GET unless it is explicitly requested using the `fields` query parameter or GET for all advanced properties is enabled. |

| Name | Type | Description |
| --- | --- | --- |
| message | string | Current state of the key manager indicating any additional steps to perform to finish the setup. This is an advanced property; there is an added cost to retrieving its value. The property is not populated for either a collection GET or an instance GET unless it is explicitly requested using the `fields` query parameter or GET for all advanced properties is enabled. |

svm

| Name | Type | Description |
| --- | --- | --- |
| _links | _links | |
| name | string | The name of the SVM. |
| uuid | string | The unique identifier of the SVM. |

volume_encryption

Indicates whether volume encryption is supported in the cluster.

| Name | Type | Description |
| --- | --- | --- |
| code | integer | Code corresponding to the status message. Returns a 0 if volume encryption is supported in all nodes of the cluster. |
| message | string | Reason for not supporting volume encryption. |
| supported | boolean | Set to true when volume encryption support is available on all nodes of the cluster. |

security_key_manager

| Name | Type | Description |
| --- | --- | --- |
| _links | _links | |

| Name | Type | Description |
| --- | --- | --- |
| external | external | Configures external key management |
| is_default_data_at_rest_encryption_disabled | boolean | Indicates whether default data-at-rest encryption is disabled in the cluster. |
| onboard | onboard | Configures onboard key management. After configuring onboard key management, save the encrypted configuration data in a safe location so that you can use it if you need to perform a manual recovery operation. |
| scope | string | Set to "svm" for interfaces owned by an SVM. Otherwise, set to "cluster". |
| status | status | Optional status information on the current state of the key manager indicating if it is fully setup or requires more action. |
| svm | svm | |
| uuid | string | |
| volume_encryption | volume_encryption | Indicates whether volume encryption is supported in the cluster. |

error_arguments

| Name | Type | Description |
| --- | --- | --- |
| code | string | Argument code |
| message | string | Message argument |

error

| Name | Type | Description |
| --- | --- | --- |
| arguments | array[error_arguments] | Message arguments |
| code | string | Error code |

| Name | Type | Description |
| --- | --- | --- |
| message | string | Error message |
| target | string | The target parameter that caused the error. |

# Create a key manager

POST `/security/key-managers`

Creates a key manager.

## Required properties

- `svm.uuid` or `svm.name` - Existing SVM in which to create a key manager.
- `external.client_certificate` - Client certificate. Required only when creating an external key manager.
- `external.server_ca_certificates` - Server CA certificates. Required only when creating an external key manager.
- `external.servers.server` - Key servers. Required only when creating an external key manager.
- `onboard.passphrase` - Cluster-wide passphrase. Required only when creating an Onboard Key Manager.
- `synchronize` - Synchronizes missing onboard keys on any node in the cluster. Required only when creating an Onboard Key Manager at the partner site of a MetroCluster configuration.

## Related ONTAP commands

- `security key-manager external enable`
- `security key-manager onboard enable`
- `security key-manager onboard sync`

## Learn more

- DOC /security/key-managers

## Parameters

| Name | Type | In | Required | Description |
| --- | --- | --- | --- | --- |
| return_records | boolean | query | False | The default is false. If set to true, the records are returned. |

## Request Body

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| external | external | Configures external key management |
| is_default_data_at_rest_encryption_disabled | boolean | Indicates whether default data-at-rest encryption is disabled in the cluster. |
| onboard | onboard | Configures onboard key management. After configuring onboard key management, save the encrypted configuration data in a safe location so that you can use it if you need to perform a manual recovery operation. |
| scope | string | Set to "svm" for interfaces owned by an SVM. Otherwise, set to "cluster". |
| status | status | Optional status information on the current state of the key manager indicating if it is fully setup or requires more action. |
| svm | svm | |
| uuid | string | |
| volume_encryption | volume_encryption | Indicates whether volume encryption is supported in the cluster. |

**Example request**


```
{
  "_links": {
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "external": {
    "client_certificate": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
    },
    "server_ca_certificates": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
    },
    "servers": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "connectivity": {
        "records": {
          "node": {
            "_links": {
              "self": {
                "href": "/api/resourcelink"
              }
            },
            "name": "node1",
            "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
          },
          "state": "not_responding"
        }
      },
      "server": "keyserver1.com:5698",
```

```
      "timeout": 60,
      "username": "admin"
    }
  },
  "onboard": {
    "existing_passphrase": "The cluster password of length 32-256 ASCII
characters.",
    "key_backup": "'--------------------------BEGIN
BACKUP--------------------------
TmV0QXBwIEtleSBCbG9iAAEAAAAEAAAAcAEAAAAAAAAxBFWWAAAAACEAAAAAAAAA
QAAAAAAAAABzDyyVAAAAALI5Jsjvy6gUxnT78KoDKXHYb6sSeraM00quOULY6BeV
n6dMFxuErCD1lbERaOQZSuaYy1p8oQHtTEfGMLZM4TYiAAAAAAAAACgAAAAAAAAA
3WTh7gAAAAAAAAAAAAAAAAIAAAAAAAgAZJEIWvdeHr5RCAvHGclo+wAAAAAAAAAA
IgAAAAAAAAoAAAAAAAAAEOTcR0AAAAAAAAAAAAAAAACAAAAAAAJAGr3tJA/LRzU
QRHwv+1aWvAAAAAAAAAACQAAAAAAAAgAAAAAAAADV1Vd/AAAAAMFM9Q229Bhp
mDaTSdqku5DCd8wG+fOZSr4bx4JT5WHvV/r5gJnXDQQAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAABOZXRBcHAgS2V5IEJsb2IA
AQAAAAMAAAAYAQAAAAAALgePkcAAAAAIgAAAAAAAAAoAAAAAAAAAEOTcR0AAAAA
AAAAAAAAAAACAAAAAAAJAGr3tJA/LRzUQRHwv+1aWvAAAAAAAAAAACIAAAAAAAAA
KAAAAAAAAACIlCHZAAAAAAAAAAAAAAAAAgAAAAAAAQCafcabsxRXMM7gxhLRrzxh
AAAAAAAAAAAkAAAAAAAAAIAAAAAAAAAA2JjQBQAAAACt4IqXcNpVggahl0axLsN4
yQjnNVKWY7mANB29O42hI7b70DTGCTaVAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAE5ldEFwcCBLZXkgQmxvYgABAAAAAwAAABgBAAAAAAAA
7sbaoQAAAAAiAAAAAAAAACgAAAAAAAAQ5NxHQAAAAAAAAAAAAAAAAAIAAAAAAAkA
ave0kD8tHNRBEfC/7Vpa8AAAAAAAAAAAIgAAAAAAAAoAAAAAAAALOHfWkAAAAA
AAAAAAAAAAACAAAAAAABAMoI9UxrHOGthQm/CB+EHdAAAAAAAAAACQAAAAAAAAA
gAAAAAAAAACnMmUtAAAAAGVk8AtPzENFgsGdsFvnmucmYrlQCsFew0HDSFKaZqK6
W8IEVzBAhPoAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
--------------------------END BACKUP--------------------------'",
    "passphrase": "The cluster password of length 32-256 ASCII
characters."
  },
  "scope": "svm",
  "status": {
    "code": 346758,
    "message": "This cluster is part of a MetroCluster configuration.
Use the REST API POST method security/key_managers/ with the
```

```
synchronize option and the same passphrase on the partner cluster
before proceeding with any key manager operations.  Failure to do so
could lead to switchover or switchback failure."
  },
  "svm": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "svm1",
    "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
  },
  "uuid": "string",
  "volume_encryption": {
    "code": 346758,
    "message": "No platform support for volume encryption in following
nodes - node1, node2."
  }
}
```


## Response

```
Status: 201, Created
```

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| num_records | integer | Number of records |
| records | array[security_key_manager] | |

**Example response**


```
{
  "_links": {
    "next": {
      "href": "/api/resourcelink"
    },
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "records": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "external": {
      "client_certificate": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
          }
        },
        "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
      },
      "server_ca_certificates": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
          }
        },
        "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
      },
      "servers": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
          }
        },
        "connectivity": {
          "records": {
            "node": {
              "_links": {
                "self": {
                  "href": "/api/resourcelink"
```

```
                }
              },
              "name": "node1",
              "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
            },
            "state": "not_responding"
          }
        },
        "server": "keyserver1.com:5698",
        "timeout": 60,
        "username": "admin"
      }
    },
    "onboard": {
      "existing_passphrase": "The cluster password of length 32-256
ASCII characters.",
      "key_backup": "'--------------------------BEGIN
BACKUP--------------------------
TmV0QXBwIEtleSBCbG9iAAEAAAAEAAAAcAEAAAAAAAAxBFWWAAAAACEAAAAAAAAA
QAAAAAAAAABzDyyVAAAAALI5Jsjvy6gUxnT78KoDKXHYb6sSeraM00quOULY6BeV
n6dMFxuErCD1lbERaOQZSuaYy1p8oQHtTEfGMLZM4TYiAAAAAAAAACgAAAAAAAAA
3WTh7gAAAAAAAAAAAAAAAAIAAAAAAAgAZJEIWvdeHr5RCAvHGclo+wAAAAAAAAAA
IgAAAAAAAAoAAAAAAAAAEOTcR0AAAAAAAAAAAAAAAACAAAAAAAJAGr3tJA/LRzU
QRHwv+1aWvAAAAAAAAAACQAAAAAAAAgAAAAAAAADV1Vd/AAAAAMFM9Q229Bhp
mDaTSdqku5DCd8wG+fOZSr4bx4JT5WHvV/r5gJnXDQQAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAABOZXRBcHAgS2V5IEJsb2IA
AQAAAAMAAAAYAQAAAAAALgePkcAAAAAIgAAAAAAAAAoAAAAAAAAAEOTcR0AAAAA
AAAAAAAAAAACAAAAAAAJAGr3tJA/LRzUQRHwv+1aWvAAAAAAAAAAACIAAAAAAAAA
KAAAAAAAAACIlCHZAAAAAAAAAAAAAAAAAgAAAAAAAQCafcabsxRXMM7gxhLRrzxh
AAAAAAAAAAAkAAAAAAAAAIAAAAAAAAAA2JjQBQAAAACt4IqXcNpVggahl0axLsN4
yQjnNVKWY7mANB29O42hI7b70DTGCTaVAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAE5ldEFwcCBLZXkgQmxvYgABAAAAAwAAABgBAAAAAAAA
7sbaoQAAAAAiAAAAAAAAACgAAAAAAAAQ5NxHQAAAAAAAAAAAAAAAAAIAAAAAAAkA
ave0kD8tHNRBEfC/7Vpa8AAAAAAAAAAAIgAAAAAAAAoAAAAAAAAALOHfWkAAAAA
AAAAAAAAAAACAAAAAAABAMoI9UxrHOGthQm/CB+EHdAAAAAAAAAAACQAAAAAAAAA
gAAAAAAAAACnMmUtAAAAAGVk8AtPzENFgsGdsFvnmucmYrlQCsFew0HDSFKaZqK6
W8IEVzBAhPoAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
```

```
----------------------------END BACKUP----------------------------'",
      "passphrase": "The cluster password of length 32-256 ASCII
characters."
    },
    "scope": "svm",
    "status": {
      "code": 346758,
      "message": "This cluster is part of a MetroCluster configuration.
Use the REST API POST method security/key_managers/ with the
synchronize option and the same passphrase on the partner cluster
before proceeding with any key manager operations.  Failure to do so
could lead to switchover or switchback failure."
    },
    "svm": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "svm1",
      "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
    },
    "uuid": "string",
    "volume_encryption": {
      "code": 346758,
      "message": "No platform support for volume encryption in
following nodes - node1, node2."
    }
  }
}
```


## Error

```
Status: Default
```

ONTAP Error Response Codes

| Error Code | Description |
|---|---|
| 65536038 | A maximum of 4 active key servers are allowed. |
| 65536214 | Failed to generate cluster key encryption key. |
| 65536216 | Failed to add cluster key encryption key. |

| Error Code | Description |
| --- | --- |
| 65536310 | Failed to setup onboard key management because the MetroCluster peer is unhealthy. |
| 65536508 | The platform does not support data at rest encryption. |
| 65536821 | The certificate is not installed. |
| 65536822 | Multitenant key management is not supported in the current cluster version. |
| 65536823 | The SVM has key manager already configured. |
| 65536824 | Multitenant key management is not supported in MetroCluster configurations. |
| 65536834 | Failed to get existing key-server details for the SVM. |
| 65536852 | Failed to query supported KMIP protocol versions. |
| 65536870 | Key management servers already configured. |
| 65536871 | Duplicate key management servers exist. |
| 65536876 | External key management requires client and server CA certificates installed and with one or more key servers provided. |
| 65536878 | External key management cannot be configured as one or more volume encryption keys of the SVM are stored in cluster key management server. |
| 65536895 | External key manager cannnot be configured since this cluster is part of a MetroCluster configuration and the partner site of this MetroCluster configuration has Onboard Key Manager configured. |
| 65536900 | Onboard key management cannot be configured because this cluster is part of a MetroCluster configuration and the partner site has the external key manager configured. |
| 65536903 | Onboard key management has failed to configure on some nodes in the cluster. Use the CLI to sync the onboard key management configuration on failed nodes. |
| 65536906 | Onboard key management has already been configured at the partner site. Use the CLI to sync the onboard key management with the same passphrase. |
| 65536907 | Onboard key management is already configured. Use the CLI to sync any nodes with onboard key management configuration. |
| 65536916 | Onboard key management is only supported for an admin SVM. |
| 65536920 | The Onboard Key Manager passphrase length is incorrect. |

| Error Code | Description |
| --- | --- |
| 65537240 | The Onboard Key Manager passphrase must be provided when performing a POST/synchronize operation. |
| 65537241 | The Onboard Key Manager existing_passphrase must not be provided when performing a POST/synchronize operation. |
| 65537244 | Unable to sync/create Onboard Key Manager on the local cluster; Onboard Key Manager is already configured on the cluster. |
| 65537245 | Unable to sync/create Onboard Key Manager on the local cluster; Onboard Key Manager is not configured on the partner cluster. |
| 65537246 | Unable to sync/create Onboard Key Manager on local cluster. This cluster is not part of a MetroCluster configuration. |
| 66060338 | Failed to establish secure connection for a key management server due to incorrect server_ca certificates. |
| 66060339 | Failed to establish secure connection for a key management server due to incorrect client certificates. |
| 66060340 | Failed to establish secure connection for a key management server due to Cryptsoft error. |
| 66060341 | Failed to establish secure connection for a key management server due to network configuration issues. |

| Name | Type | Description |
| --- | --- | --- |
| error | error | |

**Example error**

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

## Definitions

**See Definitions**

href

| Name | Type | Description |
|---|---|---|
| href | string | |

_links

| Name | Type | Description |
|---|---|---|
| self | href | |

client_certificate

Client certificate

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| uuid | string | Certificate UUID |

server_ca_certificates

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| uuid | string | Certificate UUID |

self_link

| Name | Type | Description |
|---|---|---|
| self | href | |

node

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| name | string | |
| uuid | string | |

key_server_state

The state of the key server for a specific node.

| Name | Type | Description |
| --- | --- | --- |
| node | node | |
| state | string | Key server connectivity state |

connectivity

This property returns the key server connectivity state on all nodes of the cluster. The state is returned for a node only if the connectivity is not in an available state on that node. This is an advanced property; there is an added cost to retrieving its value. The property is not populated for either a collection GET or an instance GET unless it is explicitly requested using the `fields` query parameter or GET for all advanced properties is enabled.

| Name | Type | Description |
| --- | --- | --- |
| cluster_availability | boolean | Set to true when key server connectivity state is available on all nodes of the cluster. |
| records | array[key_server_state] | An array of key server connectivity states for each node. |

key_server_readcreate

| Name | Type | Description |
| --- | --- | --- |
| _links | self_link | |
| connectivity | connectivity | This property returns the key server connectivity state on all nodes of the cluster. The state is returned for a node only if the connectivity is not in an available state on that node. This is an advanced property; there is an added cost to retrieving its value. The property is not populated for either a collection GET or an instance GET unless it is explicitly requested using the `fields` query parameter or GET for all advanced properties is enabled. |
| server | string | External key server for key management. If no port is provided, a default port of 5696 is used. |

| Name | Type | Description |
| --- | --- | --- |
| timeout | integer | I/O timeout in seconds for communicating with the key server. |
| username | string | Username credentials for connecting with the key server. |

external

Configures external key management

| Name | Type | Description |
| --- | --- | --- |
| client_certificate | client_certificate | Client certificate |
| server_ca_certificates | array[server_ca_certificates] | The UUIDs of the server CA certificates already installed in the cluster or SVM. The array of certificates are common for all the keyservers per SVM. |
| servers | array[key_server_readcreate] | The set of external key servers. |

onboard

Configures onboard key management. After configuring onboard key management, save the encrypted configuration data in a safe location so that you can use it if you need to perform a manual recovery operation.

| Name | Type | Description |
| --- | --- | --- |
| enabled | boolean | Is the onboard key manager enabled? |
| existing_passphrase | string | The cluster-wide passphrase. This is not audited. |
| key_backup | string | Backup of the onboard key manager's key hierarchy. It is required to save this backup after configuring the onboard key manager to help in the recovery of the cluster in case of catastrophic failures. |
| passphrase | string | The cluster-wide passphrase. This is not audited. |

| Name | Type | Description |
| --- | --- | --- |
| synchronize | boolean | Synchronizes missing onboard keys on any node in the cluster. If a node is added to a cluster that has onboard key management configured, the synchronize operation needs to be performed in a PATCH operation. In a MetroCluster configuration, if onboard key management is enabled on one site, then the synchronize operation needs to be run as a POST operation on the remote site providing the same passphrase. |

status

Optional status information on the current state of the key manager indicating if it is fully setup or requires more action.

| Name | Type | Description |
| --- | --- | --- |
| code | integer | Code corresponding to the status message. Returns 0 if the setup is complete. This is an advanced property; there is an added cost to retrieving its value. The property is not populated for either a collection GET or an instance GET unless it is explicitly requested using the `fields` query parameter or GET for all advanced properties is enabled. |
| message | string | Current state of the key manager indicating any additional steps to perform to finish the setup. This is an advanced property; there is an added cost to retrieving its value. The property is not populated for either a collection GET or an instance GET unless it is explicitly requested using the `fields` query parameter or GET for all advanced properties is enabled. |

svm

| Name | Type | Description |
| --- | --- | --- |
| _links | _links | |
| name | string | The name of the SVM. |
| uuid | string | The unique identifier of the SVM. |

volume_encryption

Indicates whether volume encryption is supported in the cluster.

| Name | Type | Description |
| --- | --- | --- |
| code | integer | Code corresponding to the status message. Returns a 0 if volume encryption is supported in all nodes of the cluster. |
| message | string | Reason for not supporting volume encryption. |
| supported | boolean | Set to true when volume encryption support is available on all nodes of the cluster. |

security_key_manager

| Name | Type | Description |
| --- | --- | --- |
| _links | _links | |
| external | external | Configures external key management |
| is_default_data_at_rest_encryption_disabled | boolean | Indicates whether default data-at-rest encryption is disabled in the cluster. |
| onboard | onboard | Configures onboard key management. After configuring onboard key management, save the encrypted configuration data in a safe location so that you can use it if you need to perform a manual recovery operation. |
| scope | string | Set to "svm" for interfaces owned by an SVM. Otherwise, set to "cluster". |

| Name | Type | Description |
| --- | --- | --- |
| status | status | Optional status information on the current state of the key manager indicating if it is fully setup or requires more action. |
| svm | svm | |
| uuid | string | |
| volume_encryption | volume_encryption | Indicates whether volume encryption is supported in the cluster. |

_links

| Name | Type | Description |
| --- | --- | --- |
| next | href | |
| self | href | |

error_arguments

| Name | Type | Description |
| --- | --- | --- |
| code | string | Argument code |
| message | string | Message argument |

error

| Name | Type | Description |
| --- | --- | --- |
| arguments | array[error_arguments] | Message arguments |
| code | string | Error code |
| message | string | Error message |
| target | string | The target parameter that caused the error. |

# Delete key managers

DELETE `/security/key-managers/{uuid}`

Deletes a key manager.

## Related ONTAP commands

- `security key-manager external disable`
- `security key-manager onboard disable`

## Learn more

- DOC /security/key-managers

## Parameters

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| uuid | string | path | True | |

## Response

```
Status: 200, Ok
```

## Error

```
Status: Default
```

ONTAP Error Response Codes

| Error Code | Description |
|---|---|
| 65536208 | Failed to delete the SVM Key ID. |
| 65536233 | Internal error. Deletion of km_wrapped_kdb key database has failed for onboard key management. |
| 65536234 | Internal error. Deletion of cluster_kdb key database has failed for onboard key management. |
| 65536239 | Encrypted volumes are found for the SVM. |
| 65536242 | One or more Storage Encryption devices are assigned an authentication key. |
| 65536242 | One or more Storage Encryption devices are assigned an authentication key. |
| 65536800 | Failed to lookup onboard keys. |
| 65536813 | Encrypted kernel core files found. |
| 65536817 | Failed to determine if key manager is safe to disable. |
| 65536822 | Multitenant key management is not supported in the current cluster version. |

| Error Code | Description |
| --- | --- |
| 65536827 | Failed to determine if the SVM has any encrypted volumes. |
| 65536828 | External key management is not enabled for the SVM. |
| 65536867 | Encrypted volumes are found for the SVM. |
| 196608301 | Failed to determine the type of encryption. |
| 196608305 | NAE aggregates are found in the cluster. |

| Name | Type | Description |
| --- | --- | --- |
| error | error | |

**Example error**

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

## Definitions

**See Definitions**

error_arguments

| Name | Type | Description |
|---|---|---|
| code | string | Argument code |
| message | string | Message argument |

error

| Name | Type | Description |
|---|---|---|
| arguments | array[error_arguments] | Message arguments |
| code | string | Error code |
| message | string | Error message |
| target | string | The target parameter that caused the error. |

# Retrieve key managers

GET `/security/key-managers/{uuid}`

Retrieves key managers.

## Expensive properties

There is an added cost to retrieving values for these properties. They are not included by default in GET results and must be explicitly requested using the `fields` query parameter. See DOC Requesting specific fields to learn more.

- `connectivity`
- `status.message`
- `status.code`

## Related ONTAP commands

- `security key-manager show-keystore`
- `security key-manager external show`
- `security key-manager external show-status`
- `security key-manager onboard show-backup`

## Learn more

- DOC /security/key-managers

## Parameters

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| uuid | string | path | True | Key manager UUID |
| fields | array[string] | query | False | Specify the fields to return. |

## Response

```
Status: 200, Ok
```

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| external | external | Configures external key management |
| is_default_data_at_rest_encryption_disabled | boolean | Indicates whether default data-at-rest encryption is disabled in the cluster. |
| onboard | onboard | Configures onboard key management. After configuring onboard key management, save the encrypted configuration data in a safe location so that you can use it if you need to perform a manual recovery operation. |
| scope | string | Set to "svm" for interfaces owned by an SVM. Otherwise, set to "cluster". |
| status | status | Optional status information on the current state of the key manager indicating if it is fully setup or requires more action. |
| svm | svm | |
| uuid | string | |

| Name | Type | Description |
| --- | --- | --- |
| volume_encryption | volume_encryption | Indicates whether volume encryption is supported in the cluster. |

**Example response**


```
{
  "_links": {
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "external": {
    "client_certificate": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
    },
    "server_ca_certificates": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
    },
    "servers": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "connectivity": {
        "records": {
          "node": {
            "_links": {
              "self": {
                "href": "/api/resourcelink"
              }
            },
            "name": "node1",
            "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
          },
          "state": "not_responding"
        }
      },
      "server": "keyserver1.com:5698",
```

```
      "timeout": 60,
      "username": "admin"
    }
  },
  "onboard": {
    "existing_passphrase": "The cluster password of length 32-256 ASCII
characters.",
    "key_backup": "'--------------------------BEGIN
BACKUP--------------------------
TmV0QXBwIEtleSBCbG9iAAEAAAAEAAAAcAEAAAAAAAAxBFWWAAAAACEAAAAAAAAA
QAAAAAAAAABzDyyVAAAAALI5Jsjvy6gUxnT78KoDKXHYb6sSeraM00quOULY6BeV
n6dMFxuErCD1lbERaOQZSuaYy1p8oQHtTEfGMLZM4TYiAAAAAAAAACgAAAAAAAAA
3WTh7gAAAAAAAAAAAAAAAAIAAAAAAAgAZJEIWvdeHr5RCAvHGclo+wAAAAAAAAAA
IgAAAAAAAAoAAAAAAAAAEOTcR0AAAAAAAAAAAAAAAACAAAAAAAJAGr3tJA/LRzU
QRHwv+1aWvAAAAAAAAAACQAAAAAAAAgAAAAAAAADV1Vd/AAAAAMFM9Q229Bhp
mDaTSdqku5DCd8wG+fOZSr4bx4JT5WHvV/r5gJnXDQQAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAABOZXRBcHAgS2V5IEJsb2IA
AQAAAAMAAAAYAQAAAAAALgePkcAAAAAIgAAAAAAAAAoAAAAAAAAAEOTcR0AAAAA
AAAAAAAAAAACAAAAAAAJAGr3tJA/LRzUQRHwv+1aWvAAAAAAAAAAACIAAAAAAAAA
KAAAAAAAAACIlCHZAAAAAAAAAAAAAAAAAgAAAAAAAQCafcabsxRXMM7gxhLRrzxh
AAAAAAAAAAAkAAAAAAAAAIAAAAAAAAAA2JjQBQAAAACt4IqXcNpVggahl0axLsN4
yQjnNVKWY7mANB29O42hI7b70DTGCTaVAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAE5ldEFwcCBLZXkgQmxvYgABAAAAAwAAABgBAAAAAAAA
7sbaoQAAAAAiAAAAAAAAACgAAAAAAAAQ5NxHQAAAAAAAAAAAAAAAAAIAAAAAAAkA
ave0kD8tHNRBEfC/7Vpa8AAAAAAAAAAAIgAAAAAAAAoAAAAAAAALOHfWkAAAAA
AAAAAAAAAAACAAAAAAABAMoI9UxrHOGthQm/CB+EHdAAAAAAAAAACQAAAAAAAA
gAAAAAAAACnMmUtAAAAAGVk8AtPzENFgsGdsFvnmucmYrlQCsFew0HDSFKaZqK6
W8IEVzBAhPoAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
--------------------------END BACKUP--------------------------'",
    "passphrase": "The cluster password of length 32-256 ASCII
characters."
  },
  "scope": "svm",
  "status": {
    "code": 346758,
    "message": "This cluster is part of a MetroCluster configuration.
Use the REST API POST method security/key_managers/ with the
```

```
synchronize option and the same passphrase on the partner cluster
before proceeding with any key manager operations.  Failure to do so
could lead to switchover or switchback failure."
  },
  "svm": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "svm1",
    "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
  },
  "uuid": "string",
  "volume_encryption": {
    "code": 346758,
    "message": "No platform support for volume encryption in following
nodes - node1, node2."
  }
}
```


## Error

```
Status: Default, Error
```

| Name | Type | Description |
|---|---|---|
| error | error | |

### Example error


```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

## Definitions

**See Definitions**

href

| Name | Type | Description |
|---|---|---|
| href | string | |

_links

| Name | Type | Description |
|---|---|---|
| self | href | |

client_certificate

Client certificate

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| uuid | string | Certificate UUID |

server_ca_certificates

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| uuid | string | Certificate UUID |

self_link

| Name | Type | Description |
|---|---|---|
| self | href | |

node

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| name | string | |
| uuid | string | |

key_server_state

The state of the key server for a specific node.

| Name | Type | Description |
| --- | --- | --- |
| node | node | |
| state | string | Key server connectivity state |

connectivity

This property returns the key server connectivity state on all nodes of the cluster. The state is returned for a node only if the connectivity is not in an available state on that node. This is an advanced property; there is an added cost to retrieving its value. The property is not populated for either a collection GET or an instance GET unless it is explicitly requested using the `fields` query parameter or GET for all advanced properties is enabled.

| Name | Type | Description |
| --- | --- | --- |
| cluster_availability | boolean | Set to true when key server connectivity state is available on all nodes of the cluster. |
| records | array[key_server_state] | An array of key server connectivity states for each node. |

key_server_readcreate

| Name | Type | Description |
| --- | --- | --- |
| _links | self_link | |
| connectivity | connectivity | This property returns the key server connectivity state on all nodes of the cluster. The state is returned for a node only if the connectivity is not in an available state on that node. This is an advanced property; there is an added cost to retrieving its value. The property is not populated for either a collection GET or an instance GET unless it is explicitly requested using the `fields` query parameter or GET for all advanced properties is enabled. |
| server | string | External key server for key management. If no port is provided, a default port of 5696 is used. |

| Name | Type | Description |
| --- | --- | --- |
| timeout | integer | I/O timeout in seconds for communicating with the key server. |
| username | string | Username credentials for connecting with the key server. |

external

Configures external key management

| Name | Type | Description |
| --- | --- | --- |
| client_certificate | client_certificate | Client certificate |
| server_ca_certificates | array[server_ca_certificates] | The UUIDs of the server CA certificates already installed in the cluster or SVM. The array of certificates are common for all the keyservers per SVM. |
| servers | array[key_server_readcreate] | The set of external key servers. |

onboard

Configures onboard key management. After configuring onboard key management, save the encrypted configuration data in a safe location so that you can use it if you need to perform a manual recovery operation.

| Name | Type | Description |
| --- | --- | --- |
| enabled | boolean | Is the onboard key manager enabled? |
| existing_passphrase | string | The cluster-wide passphrase. This is not audited. |
| key_backup | string | Backup of the onboard key manager's key hierarchy. It is required to save this backup after configuring the onboard key manager to help in the recovery of the cluster in case of catastrophic failures. |
| passphrase | string | The cluster-wide passphrase. This is not audited. |

| Name | Type | Description |
| --- | --- | --- |
| synchronize | boolean | Synchronizes missing onboard keys on any node in the cluster. If a node is added to a cluster that has onboard key management configured, the synchronize operation needs to be performed in a PATCH operation. In a MetroCluster configuration, if onboard key management is enabled on one site, then the synchronize operation needs to be run as a POST operation on the remote site providing the same passphrase. |

status

Optional status information on the current state of the key manager indicating if it is fully setup or requires more action.

| Name | Type | Description |
| --- | --- | --- |
| code | integer | Code corresponding to the status message. Returns 0 if the setup is complete. This is an advanced property; there is an added cost to retrieving its value. The property is not populated for either a collection GET or an instance GET unless it is explicitly requested using the `fields` query parameter or GET for all advanced properties is enabled. |
| message | string | Current state of the key manager indicating any additional steps to perform to finish the setup. This is an advanced property; there is an added cost to retrieving its value. The property is not populated for either a collection GET or an instance GET unless it is explicitly requested using the `fields` query parameter or GET for all advanced properties is enabled. |

svm

| Name | Type | Description |
| --- | --- | --- |
| _links | _links | |
| name | string | The name of the SVM. |
| uuid | string | The unique identifier of the SVM. |

volume_encryption

Indicates whether volume encryption is supported in the cluster.

| Name | Type | Description |
| --- | --- | --- |
| code | integer | Code corresponding to the status message. Returns a 0 if volume encryption is supported in all nodes of the cluster. |
| message | string | Reason for not supporting volume encryption. |
| supported | boolean | Set to true when volume encryption support is available on all nodes of the cluster. |

error_arguments

| Name | Type | Description |
| --- | --- | --- |
| code | string | Argument code |
| message | string | Message argument |

error

| Name | Type | Description |
| --- | --- | --- |
| arguments | array[error_arguments] | Message arguments |
| code | string | Error code |
| message | string | Error message |
| target | string | The target parameter that caused the error. |

# Update key managers

PATCH `/security/key-managers/{uuid}`

Updates a key manager.

## Required properties

- `onboard.existing_passphrase` - Cluster-wide passphrase. Required only when synchronizing the passphrase of the Onboard Key Manager.
- `synchronize` - Synchronizes missing Onboard Key Manager keys on any node in the cluster. Required only when synchronizing the Onboard Key Manager keys in a local cluster.

## Related ONTAP commands

- `security key-manager external modify`
- `security key-manager onboard sync`
- `security key-manager onboard update-passphrase`

## Learn more

- DOC /security/key-managers

## Parameters

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| uuid | string | path | True | Key manager UUID |

## Request Body

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| external | external | Configures external key management |
| is_default_data_at_rest_encryption_disabled | boolean | Indicates whether default data-at-rest encryption is disabled in the cluster. |

| Name | Type | Description |
| --- | --- | --- |
| onboard | onboard | Configures onboard key management. After configuring onboard key management, save the encrypted configuration data in a safe location so that you can use it if you need to perform a manual recovery operation. |
| scope | string | Set to "svm" for interfaces owned by an SVM. Otherwise, set to "cluster". |
| status | status | Optional status information on the current state of the key manager indicating if it is fully setup or requires more action. |
| svm | svm | |
| uuid | string | |
| volume_encryption | volume_encryption | Indicates whether volume encryption is supported in the cluster. |

**Example request**


```
{
  "_links": {
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "external": {
    "client_certificate": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
    },
    "server_ca_certificates": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
    },
    "servers": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "connectivity": {
        "records": {
          "node": {
            "_links": {
              "self": {
                "href": "/api/resourcelink"
              }
            },
            "name": "node1",
            "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
          },
          "state": "not_responding"
        }
      },
      "server": "keyserver1.com:5698",
```

```
      "timeout": 60,
      "username": "admin"
    }
  },
  "onboard": {
    "existing_passphrase": "The cluster password of length 32-256 ASCII
characters.",
    "key_backup": "'--------------------------BEGIN
BACKUP--------------------------
TmV0QXBwIEtleSBCbG9iAAEAAAAEAAAAcAEAAAAAAAAxBFWWAAAAACEAAAAAAAAA
QAAAAAAAAABzDyyVAAAAALI5Jsjvy6gUxnT78KoDKXHYb6sSeraM00quOULY6BeV
n6dMFxuErCD1lbERaOQZSuaYy1p8oQHtTEfGMLZM4TYiAAAAAAAAACgAAAAAAAAA
3WTh7gAAAAAAAAAAAAAAAAIAAAAAAAgAZJEIWvdeHr5RCAvHGclo+wAAAAAAAAAA
IgAAAAAAAAoAAAAAAAAAEOTcR0AAAAAAAAAAAAAAAACAAAAAAAJAGr3tJA/LRzU
QRHwv+1aWvAAAAAAAAAACQAAAAAAAAgAAAAAAAADV1Vd/AAAAAMFM9Q229Bhp
mDaTSdqku5DCd8wG+fOZSr4bx4JT5WHvV/r5gJnXDQQAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAABOZXRBcHAgS2V5IEJsb2IA
AQAAAAMAAAAYAQAAAAAALgePkcAAAAAIgAAAAAAAAAoAAAAAAAAAEOTcR0AAAAA
AAAAAAAAAAACAAAAAAAJAGr3tJA/LRzUQRHwv+1aWvAAAAAAAAAAACIAAAAAAAAA
KAAAAAAAAACIlCHZAAAAAAAAAAAAAAAAAgAAAAAAAQCafcabsxRXMM7gxhLRrzxh
AAAAAAAAAAAkAAAAAAAAAIAAAAAAAAAA2JjQBQAAAACt4IqXcNpVggahl0axLsN4
yQjnNVKWY7mANB29O42hI7b70DTGCTaVAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAE5ldEFwcCBLZXkgQmxvYgABAAAAAwAAABgBAAAAAAAA
7sbaoQAAAAAiAAAAAAAAACgAAAAAAAAAQ5NxHQAAAAAAAAAAAAAAAAAIAAAAAAAkA
ave0kD8tHNRBEfC/7Vpa8AAAAAAAAAAAIgAAAAAAAAoAAAAAAAAALOHfWkAAAAA
AAAAAAAAAAACAAAAAAABAMoI9UxrHOGthQm/CB+EHdAAAAAAAAAAACQAAAAAAAAA
gAAAAAAAAACnMmUtAAAAAGVk8AtPzENFgsGdsFvnmucmYrlQCsFew0HDSFKaZqK6
W8IEVzBAhPoAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
--------------------------END BACKUP--------------------------'",
    "passphrase": "The cluster password of length 32-256 ASCII
characters."
  },
  "scope": "svm",
  "status": {
    "code": 346758,
    "message": "This cluster is part of a MetroCluster configuration.
Use the REST API POST method security/key_managers/ with the
```

```
synchronize option and the same passphrase on the partner cluster
before proceeding with any key manager operations.  Failure to do so
could lead to switchover or switchback failure."
  },
  "svm": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "svm1",
    "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
  },
  "uuid": "string",
  "volume_encryption": {
    "code": 346758,
    "message": "No platform support for volume encryption in following
nodes - node1, node2."
  }
}
```


## Response

```
Status: 200, Ok
```

## Error

```
Status: Default
```

ONTAP Error Response Codes

| Error Code | Description |
|---|---|
| 65536139 | The existing passphrase value provided does not match the configured passphrase. |
| 65536150 | The new passphrase is same as old passphrase. |
| 65536404 | The passphrase does not match the accepted length. |
| 65536406 | The change of passphrase failed. |
| 65536407 | The passphrase update failed on some nodes. |
| 65536408 | The passphrase update failed on some nodes. |

| Error Code | Description |
| --- | --- |
| 65536802 | The passphrase does not match the accepted length in common criteria mode. |
| 65536821 | The certificate is not installed. |
| 65536822 | Multitenant key management is not supported in the current cluster version. |
| 65536828 | External key management is not enabled for the SVM. |
| 65536850 | New client certificate public or private keys are different from the existing client certificate. |
| 65536852 | Failed to query supported KMIP protocol versions. |
| 65536917 | Updating an onboard passhrase requires both new and existing cluster passphrase. |
| 65537242 | The Onboard Key Manager existing_passphrase must be provided when performing a PATCH/synchronize operation. |
| 65537243 | The Onboard Key Manager passphrase must not be provided when performing a PATCH/synchronize operation. |
| 66060338 | Failed to establish secure connection for a key management server due to incorrect server_ca certificates. |
| 66060339 | Failed to establish secure connection for a key management server due to incorrect client certificates. |
| 66060340 | Failed to establish secure connection for a key management server due to Cryptsoft error. |
| 66060341 | Failed to establish secure connection for a key management server due to network configuration issues. |

| Name | Type | Description |
| --- | --- | --- |
| error | error | |

**Example error**

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

## Definitions

**See Definitions**

href

| Name | Type | Description |
|---|---|---|
| href | string | |

_links

| Name | Type | Description |
|---|---|---|
| self | href | |

client_certificate

Client certificate

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| uuid | string | Certificate UUID |

server_ca_certificates

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| uuid | string | Certificate UUID |

self_link

| Name | Type | Description |
|---|---|---|
| self | href | |

node

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| name | string | |
| uuid | string | |

key_server_state

The state of the key server for a specific node.

| Name | Type | Description |
| --- | --- | --- |
| node | node | |
| state | string | Key server connectivity state |

connectivity

This property returns the key server connectivity state on all nodes of the cluster. The state is returned for a node only if the connectivity is not in an available state on that node. This is an advanced property; there is an added cost to retrieving its value. The property is not populated for either a collection GET or an instance GET unless it is explicitly requested using the `fields` query parameter or GET for all advanced properties is enabled.

| Name | Type | Description |
| --- | --- | --- |
| cluster_availability | boolean | Set to true when key server connectivity state is available on all nodes of the cluster. |
| records | array[key_server_state] | An array of key server connectivity states for each node. |

key_server_readcreate

| Name | Type | Description |
| --- | --- | --- |
| _links | self_link | |
| connectivity | connectivity | This property returns the key server connectivity state on all nodes of the cluster. The state is returned for a node only if the connectivity is not in an available state on that node. This is an advanced property; there is an added cost to retrieving its value. The property is not populated for either a collection GET or an instance GET unless it is explicitly requested using the `fields` query parameter or GET for all advanced properties is enabled. |
| server | string | External key server for key management. If no port is provided, a default port of 5696 is used. |

| Name | Type | Description |
| --- | --- | --- |
| timeout | integer | I/O timeout in seconds for communicating with the key server. |
| username | string | Username credentials for connecting with the key server. |

external

Configures external key management

| Name | Type | Description |
| --- | --- | --- |
| client_certificate | client_certificate | Client certificate |
| server_ca_certificates | array[server_ca_certificates] | The UUIDs of the server CA certificates already installed in the cluster or SVM. The array of certificates are common for all the keyservers per SVM. |
| servers | array[key_server_readcreate] | The set of external key servers. |

onboard

Configures onboard key management. After configuring onboard key management, save the encrypted configuration data in a safe location so that you can use it if you need to perform a manual recovery operation.

| Name | Type | Description |
| --- | --- | --- |
| enabled | boolean | Is the onboard key manager enabled? |
| existing_passphrase | string | The cluster-wide passphrase. This is not audited. |
| key_backup | string | Backup of the onboard key manager's key hierarchy. It is required to save this backup after configuring the onboard key manager to help in the recovery of the cluster in case of catastrophic failures. |
| passphrase | string | The cluster-wide passphrase. This is not audited. |

| Name | Type | Description |
| --- | --- | --- |
| synchronize | boolean | Synchronizes missing onboard keys on any node in the cluster. If a node is added to a cluster that has onboard key management configured, the synchronize operation needs to be performed in a PATCH operation. In a MetroCluster configuration, if onboard key management is enabled on one site, then the synchronize operation needs to be run as a POST operation on the remote site providing the same passphrase. |

status

Optional status information on the current state of the key manager indicating if it is fully setup or requires more action.

| Name | Type | Description |
| --- | --- | --- |
| code | integer | Code corresponding to the status message. Returns 0 if the setup is complete. This is an advanced property; there is an added cost to retrieving its value. The property is not populated for either a collection GET or an instance GET unless it is explicitly requested using the `fields` query parameter or GET for all advanced properties is enabled. |
| message | string | Current state of the key manager indicating any additional steps to perform to finish the setup. This is an advanced property; there is an added cost to retrieving its value. The property is not populated for either a collection GET or an instance GET unless it is explicitly requested using the `fields` query parameter or GET for all advanced properties is enabled. |

svm

| Name | Type | Description |
| --- | --- | --- |
| _links | _links | |
| name | string | The name of the SVM. |
| uuid | string | The unique identifier of the SVM. |

volume_encryption

Indicates whether volume encryption is supported in the cluster.

| Name | Type | Description |
| --- | --- | --- |
| code | integer | Code corresponding to the status message. Returns a 0 if volume encryption is supported in all nodes of the cluster. |
| message | string | Reason for not supporting volume encryption. |
| supported | boolean | Set to true when volume encryption support is available on all nodes of the cluster. |

security_key_manager

| Name | Type | Description |
| --- | --- | --- |
| _links | _links | |
| external | external | Configures external key management |
| is_default_data_at_rest_encryption_disabled | boolean | Indicates whether default data-at-rest encryption is disabled in the cluster. |
| onboard | onboard | Configures onboard key management. After configuring onboard key management, save the encrypted configuration data in a safe location so that you can use it if you need to perform a manual recovery operation. |
| scope | string | Set to "svm" for interfaces owned by an SVM. Otherwise, set to "cluster". |

| Name | Type | Description |
| --- | --- | --- |
| status | status | Optional status information on the current state of the key manager indicating if it is fully setup or requires more action. |
| svm | svm | |
| uuid | string | |
| volume_encryption | volume_encryption | Indicates whether volume encryption is supported in the cluster. |

error_arguments

| Name | Type | Description |
| --- | --- | --- |
| code | string | Argument code |
| message | string | Message argument |

error

| Name | Type | Description |
| --- | --- | --- |
| arguments | array[error_arguments] | Message arguments |
| code | string | Error code |
| message | string | Error message |
| target | string | The target parameter that caused the error. |

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